

Brain Activity and Decision-Making with Haptic Cues

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ABSTRACT

In this study, we investigated the relationship between behavioral and brain responses during haptic interactions in a virtual environment. Using a helicopter aerial firefighting simulation, ten participants interacted with co-located and dis-located haptic cues. Behavioral data, including reaction times were analyzed using Drift-Diffusion Model (DDM) alongside event-related potentials, focusing on N200 and P300 features. Bland-Altman analyses revealed potential agreement between DDM drift rate parameter and ERP feature slopes, suggesting that internal cognitive processes can be inferred behaviorally. The results indicate that behavioral models, such as DDM, may reduce the need for direct brain measurements, with implications for haptic interface design.

Index terms: HCI theory, concepts and models; virtual reality

1 INTRODUCTION

Computer based virtual environments (VEs) are often used to train technical skills that would be prohibitive or difficult to train in the real world. Haptic stimuli, that are physical cues related to the sense of touch, are essential in enhancing the training effectiveness and efficiency [1]. One such example of a highly technical skill requiring haptic stimuli as part of the training is the aircraft pilot training. Flight simulators are often employed to train pilots because of the cost and risk training in a real aircraft. In a flight simulator-based training session, the user of the simulator requires the feedback from a yoke or joystick to train their intuition on the behavior of the aircraft. This intuitive user interactions provide greater user immersion, and better translates a user's training to real world scenarios [1].

Despite the critical role of haptic cues, the cognitive mechanisms underlying haptic interactions remain poorly understood, primarily due to the inherent complexity of the human brain. This poses significant challenges for the design of effective haptic interfaces. One approach to glimpse into the underlying mechanisms is through examining event-related potential (ERP) features in response to a haptic cue through electroencephalography (EEG). The N200 ERP component, which is a negative voltage deflection peaking approximately 200 milliseconds post-cue onset (Fig. 1), is associated with cognitive functions such as identification of physical cues [2]. P300 is another important ERP feature. The P300 is the positive voltage deflection in ERP at approximately 300 milliseconds after a cue onset (Fig. 1). The P300's amplitude and latency is closely tied to attentional processes and the discrimination of cues [2,3]. While ERP components are

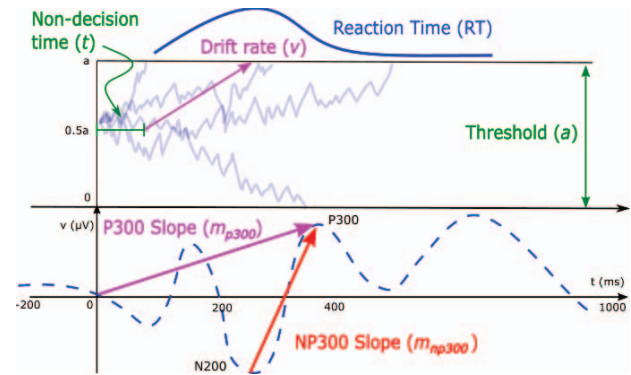


Figure 1: Example of Drift-diffusion model (DDM) and event-related potential (ERP), with drift rate (v) and slopes (m_{p300} , m_{np300}).

immensely useful in examining cognitive processes, how they link to observed user behaviors is underexplored.

One promising approach to link behavioral observations to underlying brain activities is through the Drift-Diffusion Model (DDM) [3]. Using observed reaction times (RTs) to haptic cues, DDM facilitates the decomposition of user decision-making into constituent processes described with parameters such as drift rate (v), non-decision time (t) and threshold (a) (Fig. 1). The DDM parameter of interest is v due to its association with the slope of the P300 component (i.e. the ratio of P300 amplitude and latency, m_{p300}) of ERP (Fig. 1) [3]. A previous study has also shown the potential of linking v with the NP300 slope (m_{np300}) (Fig. 1) [4]. However, whether this link holds under real world scenarios is unknown. Hence in this work we examined the relationship between m_{p300} , m_{np300} and v under a quasi-practical scenario.

2 EXPERIMENT

A helicopter aerial firefighting scenario was chosen as the quasi-practical scenario to be simulated in a VE. This training scenario involves the task of water collection from a lake via vibrotactile and haptic cues. Ten healthy volunteers (23.5 ± 3.1 years old) participated in the study. Being naïve to the purpose of the study, all of them were right-handed, with normal or corrected-to-normal vision, and 3D stereo acuity of at least 40" arc.

The 3D stereoscopic VE was developed using Unity (2019.4.28f1). The viewport of the VE scene consisted of the interior cockpit view of a helicopter in a mountainous region with a lake, as shown in Fig. 2(a). A water collector was attached via cables to the bottom of the helicopter. During the task, the user used the stylus of a haptic device (Omni; 3D Systems, USA) to control the forward and sideways movement of the helicopter, while the VE guided the helicopter along a downward curved path towards the lake surface for water collection. The stylus provided a downward force of 500mN to the user's hand when the collector reaches the water surface for 1.0s. Simultaneously, a custom kit delivered a vibrotactile cue (200Hz, 0.9G) to the user via a controller (Atmel SAM4S Xplained Pro; Atmel Corp., USA) operating two 10 mm vibrators (Precision Microdrives, UK) embedded in a Velcro wristband. The wristband was attached to the

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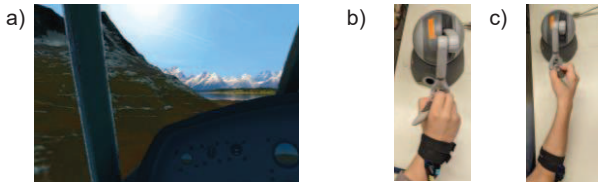


Figure 2: The setup of the VE: (a) its visual scene; (b) co-located cues; and (c) dis-located cues.

TABLE 1. CO-LOCATED AND DIS-LOCATED METRICS

Metric	Co-located (mean $\pm$ s.d.)	Dis-located (mean $\pm$ s.d.)
Workload	121 $\pm$ 25	113 $\pm$ 15
Reaction time (s)	0.479 $\pm$ 0.088	0.482 $\pm$ 0.101
Drift rate (uV/s)	0.0353 $\pm$ 0.0015	0.0358 $\pm$ 0.0014
NP300 Slope (uV/s)	0.0345 $\pm$ 0.0135	0.0326 $\pm$ 0.0107
P300 Slope (uV/s)	0.0084 $\pm$ 0.0056	0.0084 $\pm$ 0.0035

user's right hand in the co-located condition (Fig. 2(b)) or to the forearm in the dis-located condition (Fig. 2(c)). An E4 bracelet (Empatica Inc., Italy) recorded physiological signals to monitor for cybersickness. During the task, the user wore an EEG cap (Enobio-32, Neuroelectronics, Spain). Between conditions, the participants completed cybersickness and NASA TLX questionnaires [5]. Each session lasted 1 hour, with ethics approval from our institute

3 DATA ACQUISITION AND ANALYSES

During the task, objective data such as E4 signals, reaction time (RT) were logged by the VE while EEG data were collected by the EEG cap. The EEG cap acquired voltage signals as raw data at 500 Hz and preprocessed with MATLAB (R2021a) and EEGLAB (v2022.0) [6]. EEG data were filtered using a bandpass filter with passband of 1-45Hz. ERP epochs were segmented from -200ms before cue onset to 1000ms after cue onset, with baseline of each epoch within the 200ms prior to cue onset. N200 amplitudes and latencies were collected by averaging signals from Pz, CP1 and P3 electrodes; while P300 were averaged from Fz, F3, FC1, FC2 F4 electrodes. User responses outside the duration of the cues were considered incorrect and were removed. Subjective data such as user reported scores of cybersickness and cognitive workload were gathered. RTs and workload were checked for their eligibility for Student's t-test with Kolmogorov-Smirnov normality test. v 's were computed with RTs using the Hierarchical DDM plugin (v0.8) [7].

4 RESULTS AND DISCUSSIONS

None of the participants experienced cybersickness, as indicated by their cybersickness scores and physiological signals, allowing data from all participants to be included in the analyses.

RTs and workload for the two conditions are summarized in Table 1. Normality tests confirmed that these measures were normally distributed, and t-tests revealed no significant differences between conditions, consistent with findings from prior studies [4]. This supports the data's suitability for EEG analyses.

Results of v , m_{np300} and m_{p300} are presented in Table 1. Bland-Altman agreement analyses evaluated agreements between v , m_{np300} and m_{p300} (Fig. 3), with most data points within limits of agreement. Less systematic bias was present in the comparison with m_{np300} , as shown by the $m - v$ differences centered closer to 0. This suggests m_{np300} might better correlate with v . However, v and m_{p300} were more closely clustered, indicating v might be better reflect P300-related brain processes. Linear patterns in Fig. 3 implies the agreement is imperfect and may depend on the task. However, larger sample sizes are necessary for confirmation. Overall, the agreement between v , m_{np300} and m_{p300} suggests that aspects of the

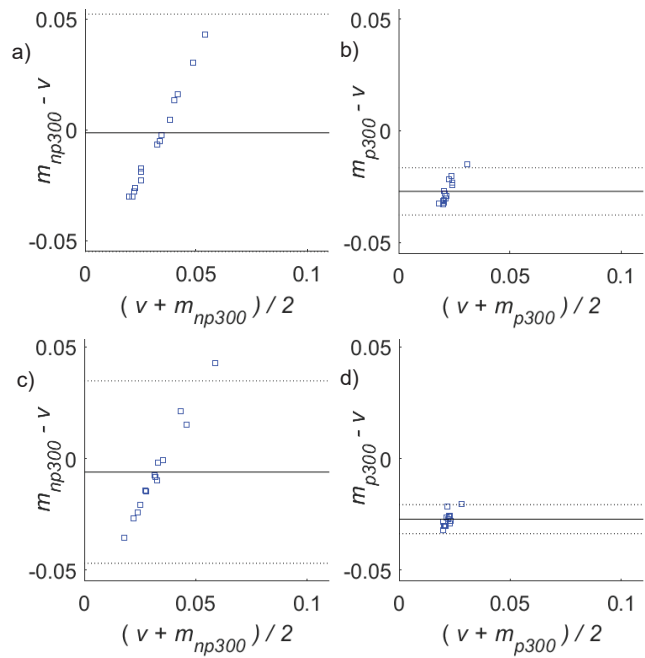


Figure 3: Bland-Altman plots between v and: a) m_{np300} with co-located cues; b) m_{p300} with co-located cues; c) m_{np300} with dis-located cues; and d) m_{p300} with dis-located cues.

internal brain process can be inferred from behavioral analyses with DDM. Furthermore, the similarities between co- and dis-located cues indicates those are both viable for presenting cues to users to address real-world equipment constraints. Overall, the findings indicate a potential to reduce the need to measure brain responses, with possibility of applying to other tasks involving curved paths.

5 CONCLUSION

v , m_{np300} and m_{p300} presented feasible links between a behavioral model and the underlying brain responses to haptic cues. Further work to examine this link under different conditions is required.

6 ACKNOWLEDGMENTS

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